STACKS, DERIVED CATEGORIES OF SPACES: DIRECT-IMAGE-COHERATOR

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2713-2734 of sources/stacks-spaces-perfect/pinned-original-source.tex; proof span is lines 2736-2813, the complete author proof (76 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 18 of queue_spaceperfect.json (pb=3409, stmt 2713-2734, proof 2736-2813).

Editorial note. stacks-spaces-perfect wave sp01 (2026-10-14): fresh adjudication accept per lane-s3/sp01_judgments; pool per Chief 12:4x unfreeze (reserve-pool workflow, six-check clean); spaces-perfect.tex 126 with-proof rows, 70 b107-occupied labels avoided, 56 free rows judged 32 accept / 24 reject.

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-direct-image-coherator' also exists in stacks-perfect — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 08BJ: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 2813–2813. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08GY: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 2339–2353. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08GQ: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1643–1662. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08H1: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 2689–2700. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08GR: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1677–1700. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08GB: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 2813–2813. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03BS: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 2813–2813. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme and let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume X and Y are quasi-compact and have affine diagonal over* $\mathbf{Z}$ *(as in Properties of Spaces, Definition 03BS). Then, denoting*

$$\Phi : D(QCoh(\mathcal{O}_X)) \to D(QCoh(\mathcal{O}_Y))$$

the right derived functor of $f_* : QCoh(\mathcal{O}_X) \to QCoh(\mathcal{O}_Y)$ the diagram

$$\begin{array}{ccc} D(QCoh(\mathcal{O}_X)) & \longrightarrow & D_{QCoh}(\mathcal{O}_X) \\ \downarrow{\scriptstyle \Phi} & & \downarrow{\scriptstyle Rf_*} \\ D(QCoh(\mathcal{O}_Y)) & \longrightarrow & D_{QCoh}(\mathcal{O}_Y) \end{array}$$

is commutative.

Proof. Observe that the horizontal arrows in the diagram are equivalences of categories by Proposition 08H1. Hence we can identify these categories (and similarly for other quasi-compact algebraic spaces with affine diagonal) and then the statement of the lemma is that the canonical map $\Phi(K) \to Rf_*(K)$ is an isomorphism for all K in $D(QCoh(\mathcal{O}_X))$. Note that if $K_1 \to K_2 \to K_3 \to K_1[1]$ is a distinguished triangle in $D(QCoh(\mathcal{O}_X))$ and the statement is true for two-out-of-three, then it is true for the third.

Let $\mathcal{B} \subset \mathrm{Ob}(X_{spaces,étale})$ be the set of objects which are quasi-compact and have affine diagonal. For $U \in \mathcal{B}$ and any morphism $g : U \to Z$ where Z is a quasi-compact algebraic space over S with affine diagonal, denote

$$\Phi_g : D(QCoh(\mathcal{O}_U)) \to D(QCoh(\mathcal{O}_Z))$$

the derived extension of g_*. Let $P(U) =$ "for any K in $D(QCoh(\mathcal{O}_U))$ and any $g : U \to Z$ as above the map $\Phi_g(K) \to Rg_*K$ is an isomorphism". By Remark 08GR conditions (1), (2), and (3)(a) of Lemma 08GQ hold and we are left with proving (3)(b) and (4).

Checking condition (3)(b). Let U be an affine scheme étale over X. Let $g : U \to Z$ be as above. Since the diagonal of Z is affine the morphism $g : U \to Z$ is affine (Morphisms of Spaces, Lemma 08GB). Hence $P(U)$ holds by Lemma 08GY.

Checking condition (4). Let $(U \subset W, V \to W)$ be an elementary distinguished square in $X_{spaces,étale}$ with U, W, V in $\mathcal{B}$ and V affine. Assume that P holds for U, V, and $U \times_W V$. We have to show that P holds for W. Let $g : W \to Z$ be

a morphism to a quasi-compact algebraic space with affine diagonal. Let K be an object of $D(QCoh(\mathcal{O}_W))$. Consider the distinguished triangle

$$K \to Rj_{U,*}K|_U \oplus Rj_{V,*}K|_V \to Rj_{U\times_W V,*}K|_{U\times_W V} \to K[1]$$

in $D(\mathcal{O}_W)$. By the two-out-of-three property mentioned above, it suffices to show that $\Phi_g(Rj_{U,*}K|_U) \to Rg_*(Rj_{U,*}K|_U)$ is an isomorphism and similarly for V and $U \times_W V$. This is discussed in the next paragraph.

Let $j : U \to W$ be a morphism $X_{spaces,étale}$ with U, W in $\mathcal{B}$ and P holds for U. Let $g : W \to Z$ be a morphism to a quasi-compact algebraic space with affine diagonal. To finish the proof we have to show that $\Phi_g(Rj_*K) \to Rg_*(Rj_*K)$ is an isomorphism for any K in $D(QCoh(\mathcal{O}_U))$. Let $\mathcal{I}^\bullet$ be a K-injective complex in $QCoh(\mathcal{O}_U)$ representing K. From $P(U)$ applied to j we see that $j_*\mathcal{I}^\bullet$ represents Rj_*K. Since $j_* : QCoh(\mathcal{O}_U) \to QCoh(\mathcal{O}_X)$ has an exact left adjoint $j^* : QCoh(\mathcal{O}_X) \to QCoh(\mathcal{O}_U)$ we see that $j_*\mathcal{I}^\bullet$ is a K-injective complex in $QCoh(\mathcal{O}_W)$, see Derived Categories, Lemma 08BJ. Hence $\Phi_g(Rj_*K)$ is represented by $g_*j_*\mathcal{I}^\bullet = (g \circ j)_*\mathcal{I}^\bullet$. By $P(U)$ applied to $g \circ j$ we see that this represents $R_{g\circ j,*}(K) = Rg_*(Rj_*K)$. This finishes the proof. □